

PROGRAM 26

AIM

To implement the Best First Search algorithm to find a path from a starting node to a goal node using heuristic values.

ALGORITHM

1. Start the program.
2. Define the graph and heuristic values for each node.
3. Select the starting node and goal node.
4. Add the starting node to the open list.
5. Select the node with the lowest heuristic value.
6. Remove the selected node from the open list and add it to the visited list.
7. Check whether the selected node is the goal node.
8. If it is the goal node, display the path.
9. Otherwise, add its unvisited neighbouring nodes to the open list.
10. Repeat the process until the goal node is reached or no nodes remain.
11. Stop the program.

PROGRAM

edge(a,b).

edge(a,c).

edge(b,d).

edge(b,e).

edge(c,f).

edge(d,g).

edge(e,g).

edge(f,g).

heuristic(a,6).

heuristic(b,4).

heuristic(c,5).

heuristic(d,2).

heuristic(e,3).

heuristic(f,4).

```

heuristic(g,0).
best_first(Start, Goal, Path) :-
    search([Start], Goal, [Start], RevPath),
    reverse(RevPath, Path).
search([Goal|_], Goal, _, [Goal]).
search([Current|Rest], Goal, Visited, Path) :-
    findall(H-Next,
        (edge(Current, Next),
         \+ member(Next, Visited),
         heuristic(Next, H)),
        Children),
    sort(Children, Sorted),
    add_nodes(Sorted, Rest, NewOpen),
    NewOpen = [Next|_],
    search(NewOpen, Goal, [Next|Visited], Path1),
    Path = [Current|Path1].
add_nodes([], Open, Open).
add_nodes([_ -Node|T], Open, NewOpen) :-
    add_nodes(T, Open, Temp),
    insert_by_heuristic(Node, Temp, NewOpen).
insert_by_heuristic(Node, [], [Node]).
insert_by_heuristic(Node, [H|T], [Node,H|T]) :-
    heuristic(Node, HN),
    heuristic(H, HH),
    HN =< HH.
insert_by_heuristic(Node, [H|T], [H|R]) :-
    heuristic(Node, HN),
    heuristic(H, HH),
    HN > HH,
    insert_by_heuristic(Node, T, R).

```

OUTPUT

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45 ?- consult(['c:/Users/Hema Neeпа/OneDrive/Documents/Best_Search.pl']).  
true.  
  
46 ?- best_first(a, g, Path).  
Path = [g, d, b, a] .
```

RESULT

The Best First Search algorithm was successfully implemented, and a path from the starting node A to the goal node G was obtained using heuristic values.